

ME 528: Thermal Management of Electronics

Course Code: ME 528

Course Name: Thermal Management of Electronics

L-T-P-C: 3-0-0-3

Prerequisites: ME 303: Heat Transfer

Intended for: DE for ME/ FE for others

Mutual Exclusion: None

Approval: 61st BoA

Preamble

The rapid miniaturization of devices and increasing power densities in modern electronics have made thermal management a critical design challenge. Efficient cooling not only ensures reliability and performance but also extends the operational lifespan of systems ranging from consumer electronics to high-power electric vehicles and aerospace systems. This course introduces students to the fundamentals and advanced concepts of thermal management in electronics, building upon the foundation of heat transfer. Topics will cover conduction, convection, and radiation in electronic systems, cooling technologies such as heat sinks, heat pipes, phase-change materials, microchannels, and immersion cooling. Through case studies, simulations, and problem-solving, students will gain the knowledge and tools to design, analyse, and optimize thermal solutions for next-generation electronics.

Course Modules with quantitative lecture hours

- **Module 1: Introduction to Thermal Management:** Importance of thermal management: reliability, performance, safety, Heat generation in semiconductor devices and power electronics, Failure mechanisms due to overheating, Thermal specifications of microelectronics packaging, Thermal design considerations and key metrics (2 Hours)
- **Module 2: Fundamentals of Heat Transfer in Electronics Cooling:** Heat transfer mechanisms, Conduction: thermal resistance networks, spreading resistance, contact resistance, thermal interface materials, Convection: natural and forced convection in electronics cooling, concept of boundary layer, laminar and turbulent flows, Radiation in compact enclosures, Transient thermal analysis (7 Hours)
- **Module 3: Cooling Technologies for Electronics:** Air cooling: fins and heat sinks, design of heat sinks, fans- performance curves, fin optimization and material selection, Flow management in Enclosures and Racks, baffles, flow straighteners, Liquid cooling: cold plates fundamentals and material selection, Microchannel heat sinks: pressure drop considerations, pump selection, applications, jet impingement, Phase-change based cooling: heat pipes, vapor chambers, boiling heat transfer, Immersion cooling and dielectric fluids, Comparative analysis of cooling methods (16 Hours)
- **Module 4: Advanced Thermal Management Approaches:** Thermoelectric cooling, Phase change materials (PCMs) for thermal storage, Additively manufactured heat exchangers, Emerging technologies: TPMS structures, nanofluids, hybrid cooling systems, Reliability and long-term performance considerations (7 Hours)
- **Module 5: System-Level Thermal Management:** Thermal management in mobile phones, data centers and high-performance computing, Thermal design in EV batteries and power electronics, Compact electronic packages and 3D integration, Co-design of electrical, mechanical, and thermal systems (6 Hours)

- **Module 6: Experimental techniques and modelling tools:** Thermal modeling tools and methods (analytical, CFD, FEM), Thermal characterization and experimental methods, IR imaging, Experiments on heat sinks and heat pipes (4 Hours)

Text Books

1. Shabany, Younes. *Heat transfer: thermal management of electronics*. CRC press, 2009.
2. Yeh, L. T. *Thermal management of microelectronic equipment: heat transfer theory, analysis methods, and design practices*. ASME Press, 2016.

References

1. Incropera, F. P. (1999). *Liquid cooling of electronic devices by single-phase convection*. Wiley.
2. Bejan, A. (2013). *Convection heat transfer*. John wiley & sons.
3. Incropera, Frank P., et al. *Fundamentals of heat and mass transfer*. Wiley, 2018.
4. Çengel, Yunus A. *Heat transfer: a practical approach*. McGraw Hill, 2002.